Youssef Ashraf

LinkedIn | GitHub | Portfolio | yousefashraf970@gmail.com | (+20) 01154202976

SUMMARY

Computer science professional passionate about deep learning and machine learning, with expertise in engineering AI-powered solutions. Experienced in building conversational AI systems, NLP pipelines, and deep learning models. Eager to contribute to innovative data-driven projects and continue deepening expertise in AI and data analysis.

EDUCATION

B.Sc. in Computer Science – Major in AI and Machine Learning **2022 – 2026**
Benha National University | GPA: 3.76 / 4.0 | Class Rank: 4th, AI and Machine Learning Major

WORK EXPERIENCE

Co-Head, Machine Learning Track **Feb 2026 – May 2026**
Google Developer Groups (GDG)

- Co-led the Machine Learning track, delivering technical sessions to students from multiple universities.
- Mentored members in machine learning fundamentals and practical applications while supporting hands-on workshops and collaborative learning initiatives.

Cloud Services Trainee **Jun 2025 – Jul 2025**
National Telecommunication Institute (NTI)

- Hands-on with 8 core AWS services: EC2, S3, IAM, VPC, Lambda, RDS, CloudWatch.
- Designed secure, scalable architectures aligned with Solutions Architect best practices.
- Reduced projected cloud costs by 20% through optimized IAM policies and networking.

Computer Vision Intern **Jul 2025 – Sep 2025**
National Academy of IT for Persons with Disabilities (NAID) — New Capital · On-Site

- Contributed to a computer vision project for dyslexia detection; processed and analyzed 500+ text samples to improve reading assistance accuracy.
- Collaborated with a team of 6 researchers; contributed to 2 technical presentations on AI applications in accessibility, achieving 85% model accuracy in preliminary testing.

PROJECTS

BNUConnect – AI-Powered Academic Advisor Chatbot — Graduation Project **2026**
Private Repo

- Engineered a production-grade AI academic advisor chatbot for Benha National University, deployed as a REST API (FastAPI + Uvicorn) integrated into the BNUConnect mobile app, serving students across 3 program tracks (AI & ML, Software Development, Data Science).
- Built a stateful LangGraph ReAct agent with a custom Judging Loop, orchestrating 12 LangChain tools across Neo4j knowledge graph (course catalog, prerequisites, program tracks), Supabase (student profiles, chat history), and a Pinecone RAG pipeline for university bylaw Q&A — achieving accurate multi-hop reasoning over the full academic domain.
- Designed a 5-step NLP preprocessing pipeline (reference resolution, entity extraction, fuzzy course name mapping, track normalization, query rewriting) that resolves abbreviations and pronouns before the agent runs, eliminating wasted tool calls and handling ambiguous queries via interactive disambiguation sessions.
- Implemented a RAG pipeline using HuggingFace multilingual-e5-large embeddings and Pinecone vector store (bnu-bylaws index) for bylaw/regulation Q&A, and built an interactive multi-turn course planner bridged from a CLI function to the chatbot via a thread I/O bridge with 90-second timeout handling.

Stack: LangGraph, LangChain, FastAPI, Groq, Neo4j Aura, Supabase (PostgreSQL), Pinecone, HuggingFace, Python, Docker.

BNU Admission Chatbot – Multilingual RAG Assistant **2026**
Private Repo

- Built a multilingual RAG chatbot answering admission queries in Arabic (MSA & Egyptian), Franco-Arabic, and English, using hybrid dense (Pinecone) + sparse (BM25) retrieval with reciprocal rank fusion, plus an intent router that skips LLM calls for greetings, FAQs, and out-of-scope messages.
- Automated knowledge-base upkeep with a Google Drive sync job detecting new/modified/deleted source PDFs every 10 minutes, shipped as a FastAPI service with an RTL/LTR-aware frontend and Docker deployment.

Stack: FastAPI, Groq, Pinecone, BM25, Cohere, HuggingFace, Google Drive API, Docker.

Pix2Pix – Image-to-Image Translation with Conditional GANs **2025**

Public Repo ↗

- Implemented the Pix2Pix conditional GAN framework (U-Net generator with skip connections + PatchGAN discriminator) in TensorFlow/Keras to translate architectural line-drawing schematics into photorealistic building facades on the 606-image CMP Facades dataset.
- Trained the custom model for 600 epochs (Adam, lr 5e-5, L1 λ=500, instance normalization) on a Tesla T4 GPU, outperforming a pretrained PyTorch baseline across MAE (0.145 vs. 0.183), PSNR (22.34 dB vs. 12.62 dB), and SSIM (0.71 vs. 0.25).
- Built quantitative model-comparison tooling and an interactive Gradio web app for real-time schematic-to-facade inference with public URL sharing.

Stack: TensorFlow, Keras, PyTorch, U-Net, PatchGAN, Gradio, NumPy.

Video Action Recognition on UCF50 **2025**

Public Repo ↗

- Trained a spatio-temporal TensorFlow/Keras classifier to recognize 50 human action categories from the UCF50 dataset, sampling, resizing, and normalizing 20 frames per clip for model input.
- Deployed an interactive Gradio app that accepts uploaded videos, transcodes them with FFmpeg, and returns top-3 predicted actions with confidence scores.

Stack: TensorFlow, Keras, OpenCV, Gradio, FFmpeg, NumPy.

Twitter Sentiment Analysis with Apache Spark & Docker **2025**

Public Repo ↗

- Built a distributed sentiment analysis pipeline on a 3-node (1 master + 2 workers) Dockerized Apache Spark cluster, extracting 20,000 TF-IDF features and training a Logistic Regression classifier on ~75K tweets to label sentiment as Positive, Negative, Neutral, or Irrelevant (91.7% training / 80.2% validation accuracy).
- Served the model through a Flask web app with a REST endpoint for real-time predictions and per-class confidence scores.

Stack: Apache Spark, PySpark MLlib, Flask, Docker Compose, Logistic Regression, TF-IDF.

ACHIEVEMENTS

- **Clash of Codes (HTI AI Hackathon) — 1st Place, NLP Track:** Placed 1st on the NLP track for a multilingual (Arabic, English, Franco-Arabic) RAG chatbot built over historical data under competition conditions.
- **ECPC (Egyptian Collegiate Programming Contest) — Finalist:** Sole qualifying team from Benha National University; solved 4 problems as a 3-person team to advance from the qualification round to the ECPC finals (ACPC qualifier, under ICPC rules).

CERTIFICATIONS

- Co-Head, Machine Learning Track — Google Developer Groups (GDG)
- Cloud Architect — National Telecommunication Institute (NTI)
- Machine Learning Specialization — Andrew Ng (Coursera)
- Deep Learning Specialization — Andrew Ng (Coursera)
- Hugging Face Agents Course
- Clash of Codes — 1st Place, NLP Track (Higher Technical Institute)
- Neo4j & Generative AI Fundamentals — Neo4j GraphAcademy
- Using Neo4j with LangChain — Neo4j GraphAcademy
- Building Knowledge Graphs with LLMs — Neo4j GraphAcademy

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C++ (Intermediate)
AI / ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn
LLM & Agentic AI: LangChain, LangGraph, CrewAI, Smolagents, LlamaIndex, RAG, Prompt Engineering, Fine-tuning, Ollama, vLLM
MLOps & Observability: LangSmith, LangFuse, Weights & Biases, Groq
Deep Learning Techniques: CNNs, RNNs, LSTMs, Transformers, Transfer Learning, LLMs
Orchestration & Deployment: FastAPI, Gradio, Streamlit, Docker, Kubernetes (EKS), GitHub Actions, GitHub
Databases: MySQL, MongoDB, Neo4j, Supabase (PostgreSQL), Elasticsearch, Redis
Vector Stores: Pinecone, FAISS, Chroma
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Statistical Analysis, Feature Engineering

LANGUAGES

Arabic – Native | English – Advanced